Does the Order of Training Data Influence Optimization?

Niels Enkaoua, Rémi Germi, Mohamed Sami Ghrab
EPFL
Optimization for Machine Learning Mini-Project, Spring 2026

***Abstract*—We study how training-data order changes stochastic gradient descent when the dataset, model, and hyperparameters are fixed. Using Fashion-MNIST, a small CNN, momentum SGD, 10 epochs, and seeds 0, 1, and 2, we compare two IID orders, two curricula, and two label-based non-IID streams. Random reshuffling gives the best final accuracy ($89.47 \pm 0.68\%$), fixed random order is slightly worse, and hard-to-easy curriculum clearly outperforms easy-to-hard. The two label-based orders collapse to chance accuracy at the reference learning rate. The recorded diagnostics show why: sorted label streams create highly aligned or non-finite update sequences, degenerate single-class predictors, and a much narrower stability region. Lowering the learning rate partially rescues training but not balanced generalization. Two further controls show the failure is general but its severity is not: AdamW partially rescues class-blocked streams yet keeps a narrow stability region, while a convex logistic-regression model degrades under the same orders but never collapses to chance. Data order therefore changes both SGD stability and the quality of the final classifier, and its catastrophic mode is specific to non-convex training.**

I. Introduction

Mini-batch SGD is usually analyzed under an approximately IID sampling assumption, which random reshuffling approximates in practice [1], [2]. If the training stream becomes structured, however, consecutive updates need not explore diverse directions: momentum can instead accumulate a persistent bias. We test that mechanism directly by varying only the order of Fashion-MNIST examples while keeping the model, optimizer, and data split fixed. The central question is whether ordering changes both optimization stability and the final multi-class classifier.

Prior work establishes *why* random reshuffling outperforms with-replacement sampling theoretically [2] and studies hand-designed curricula [3]; our experiments add the complementary empirical view: how badly, and through which mechanism, optimization fails when the IID assumption is broken, and whether that failure depends on the optimizer (SGD vs. AdamW [4]) and on the model class (non-convex CNN vs. convex logistic regression).

II. Experimental Setup

We train a compact CNN on Fashion-MNIST [5] with SGD, momentum 0.9, weight decay 5×10^{-4}, batch size 128, and reference learning rate 0.05. All main runs use 10 epochs, seeds 0, 1, and 2, and a stratified subset of 20,000 training examples. We compare `random`, `fixed_random`, `curriculum_easy`, `curriculum_hard`, `label_sorted`, and `label_block_random`. For each run we record epoch-wise accuracy and loss, gradient cosine similarity between consecutive steps, per-class test accuracy, forgetting, and final prediction distributions. Two control studies repeat key orderings with AdamW (learning rates 10^{-3} and 10^{-2}) and with a convex logistic-regression model (SGD, learning rate 0.05).

TABLE I
Training-data orders used in the study.

Ordering	Construction rule
`random`	Fresh random permutation at every epoch.
`fixed_random`	One random permutation reused for all epochs.
`curriculum_easy`	Samples sorted from easy to hard once, then reused.
`curriculum_hard`	Samples sorted from hard to easy once, then reused.
`label_sorted`	Samples fully grouped by class label in a fixed order.
`label_block_random`	Class-homogeneous blocks with reshuffled block order across epochs.

III. Results

A. Global Results

Figure 1 shows the overall ranking. Random reshuffling is the strongest baseline at $89.47 \pm 0.68\%$, followed by fixed random order at $88.46 \pm 0.49\%$. Among deterministic non-IID schedules, hard-to-easy curriculum reaches $84.53 \pm 0.93\%$ and clearly dominates easy-to-hard at $70.71 \pm 2.74\%$. The two label-based orders collapse to exactly 10% test accuracy, which is chance level for this balanced ten-class task. The main qualitative split is therefore between trainable orders (`random`, `fixed_random`, and both curricula) and class-separated streams, which fail completely at the reference step size.

B. Failure Analysis of Label-Based Orders

The failure modes are not just low-accuracy outcomes; they are unstable optimization trajectories. Figure 2 shows that `label_sorted` predicts class 0 for every test image in all three seeds, while `label_block_random` collapses to class 0 for seeds 0 and 1 and to class 1 for seed 2. The cosine traces reveal why. For `label_sorted`, only 13, 18, and 11 finite step-to-step gradient cosines remain for seeds 0, 1, and 2, with last finite steps 64, 80, and 48 out of 1570 total

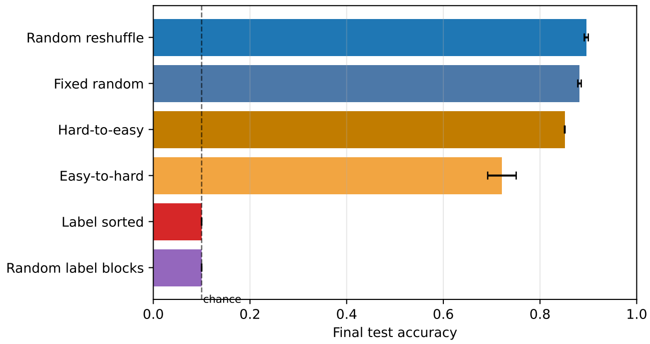

Fig. 1. Final test accuracy at learning rate 0.05. Error bars show one standard deviation across seeds 0, 1, and 2. Random reshuffling is best; both label-based orders collapse to chance.

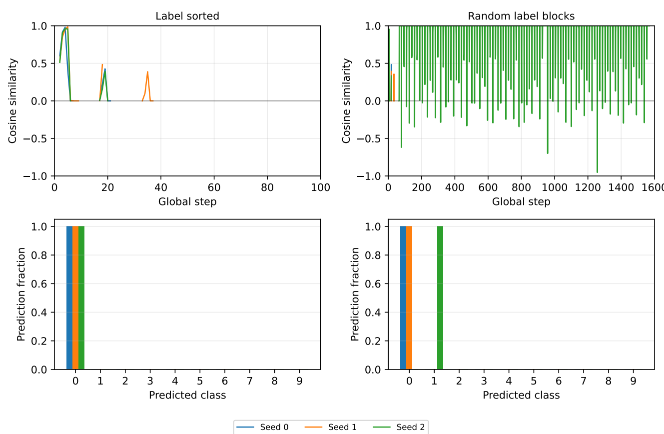

Fig. 2. Failure diagnostics at learning rate 0.05 for the two pathological orders. Top row: gradient cosine similarity per seed; lines terminate when finite values disappear. Bottom row: final predicted-class distributions per seed. `label_sorted` collapses identically across seeds, whereas `label_block_random` is seed-dependent numerically but still converges to single-class predictors.

pairs. For `label_block_random`, seeds 0 and 1 fail almost immediately (19 and 48 finite steps), while seed 2 stays finite to the end but still converges to a degenerate one-class predictor. This seed-dependent split shows that blockwise label order is numerically less brittle than fully sorted order, yet just as poor functionally.

Appendix A shows the same pattern across epochs and learning rates. At 0.05 and 0.1, both label-based orders remain at $10.00 \pm 0.00\%$. Lowering the learning rate to 0.001 partially rescues optimization, raising `label_sorted` to $31.95 \pm 8.42\%$ and `label_block_random` to $40.00 \pm 5.94\%$. However, the rescue is incomplete and still structurally biased: the dominant predicted class keeps 36–53% of the test set for `label_sorted` and 30–40% for `label_block_random`, with only five to seven classes receiving any predictions. Thus a smaller step size softens collapse but does not restore balanced multi-class learning. Appendix B confirms this at the class level and is consistent with catastrophic forgetting under sequentially biased streams [6].

C. *Stable Methods*

The four trainable orders remain finite for essentially the whole run: each keeps valid gradient-cosine values on more than 99.9% of recorded step pairs. The two IID baselines also have the weakest directional alignment, with mean cosine similarity 0.200 for `random` and 0.201 for `fixed_random`. The curriculum variants are more correlated (0.249 for `curriculum_easy`, 0.259 for `curriculum_hard`), which is consistent with their lower final accuracy and with the idea that deterministic structure reduces gradient diversity even when it does not cause outright collapse.

The stable methods differ mainly in class balance and retention. Appendix C shows that `random` and `fixed_random` keep the most even class-wise behavior: at epoch 10, their weakest-class accuracies are 0.680 and 0.667, respectively. For `curriculum_hard`, the weakest class remains much lower at 0.447, but this is still substantially better than `curriculum_easy`, whose weakest class reaches only 0.273. The forgetting diagnostic tells the same story. Final mean forgetting is small for `random` (0.0467), `fixed_random` (0.0427), and `curriculum_hard` (0.0253), but rises to 0.123 for `curriculum_easy`. Among deterministic curricula, the order of difficulty therefore matters: hard-to-easy remains clearly more stable and better balanced than easy-to-hard.

D. *Robustness Across Optimizers and Model Classes*

Two control studies test whether the ordering effects above are specific to momentum SGD on a non-convex model (Table II). First, AdamW at its standard learning rate 10^{-3} reproduces the ordering hierarchy and *partially* rescues the class-blocked stream: `label_block_random` reaches $40.6 \pm 13.3\%$ instead of collapsing to 10% as under SGD at the reference step size, suggesting that adaptive per-coordinate step sizes dampen the bias accumulated over class-homogeneous updates. The rescue is fragile: at learning rate 10^{-2}, AdamW collapses to chance not only on `label_block_random` but also on `curriculum_hard`, which SGD trains without difficulty, and `fixed_random` becomes seed-dependent ($61.1 \pm 44.2\%$). The narrow stability region under structured streams is therefore not an artifact of SGD; for the adaptive optimizer it is, if anything, narrower. Second, the convex logistic-regression model never collapses: its worst order (`label_sorted`, 52.9%) stays far above chance, although ordering still costs up to 28 accuracy points relative to random reshuffling ($80.9 \pm 0.3\%$). The catastrophic single-class failure mode therefore requires non-convexity; on a convex objective, structured orders merely slow and bias optimization, in line with incremental-gradient theory [1], [2].

TABLE II

FINAL TEST ACCURACY (%, MEAN $\pm$ STD OVER SEEDS 0–2) FOR THE OPTIMIZER AND MODEL-CLASS CONTROLS. CNN/SGD AT THE REFERENCE LEARNING RATE 0.05 IS THE MAIN EXPERIMENT (FIG. 1). DASHES DENOTE ORDERINGS NOT INCLUDED IN THE CONTROL RUNS.

	CNN, AdamW		Logistic, SGD
Ordering	lr 10^{-3}	lr 10^{-2}	lr 0.05
`random`	89.6 ± 0.4	87.1 ± 0.8	80.9 ± 0.3
`fixed_random`	88.6 ± 0.2	61.1 ± 44.2	81.0 ± 1.6
`curriculum_easy`	–	–	58.4 ± 0.8
`curriculum_hard`	86.3 ± 0.3	10.0 ± 0.0	79.0 ± 0.0
`label_sorted`	–	–	52.9 ± 0.0
`label_block_random`	40.6 ± 13.3	10.0 ± 0.0	58.0 ± 11.1

IV. CONCLUSION

Data order changes SGD in a mechanically meaningful way. Random reshuffling gives the best and most robust behavior, fixed random order is consistently weaker, and curriculum design matters even when the stream remains trainable. The strongest effect appears for class-separated streams, which produce degenerate single-class predictors, non-finite gradient diagnostics, and a much narrower learning-rate stability region. Lowering the step size helps, but it does not remove the structural bias introduced by non-IID ordering. The control studies sharpen this picture: adaptive step sizes (AdamW) mitigate but do not repair the damage, and a convex model degrades gracefully where the CNN collapses, identifying the catastrophic mode as a genuinely non-convex phenomenon. In this controlled setting, data order is therefore not a benign implementation detail but a direct control knob on optimization stability and generalization.

a) Reproducibility: The code, all stored experiment outputs, and the complete commands to reproduce every figure and table in this report are available in the accompanying repository; see its `README` for step-by-step instructions.

APPENDIX A
ADDITIONAL ACCURACY AND LEARNING-RATE CURVES

This appendix reports the epoch-wise accuracy trajectories for all six orders and the learning-rate sensitivity of the two failing orders (Figs. 3 and 4).

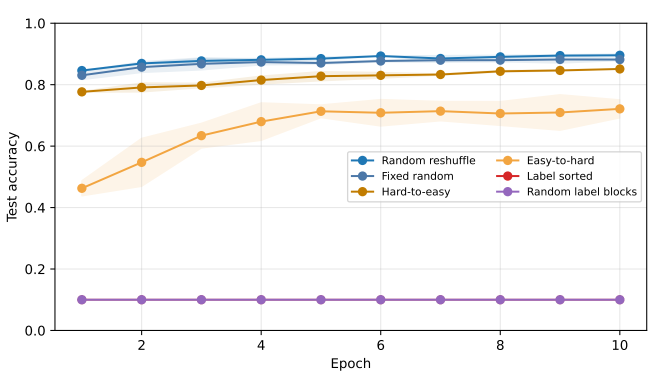

Fig. 3. Test-accuracy trajectories for all six orders at the reference learning rate 0.05. The two label-based orders collapse from the first epoch.

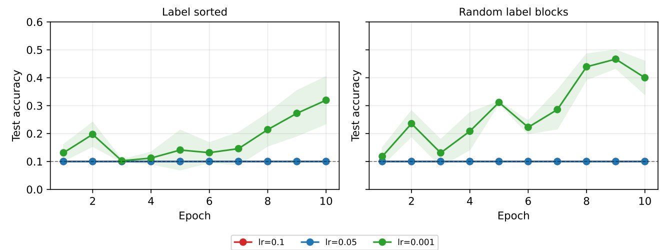

Fig. 4. Learning-rate sensitivity of the failing orders. Lowering the learning rate to 0.001 partially rescues test accuracy but does not approach the reshuffled baseline.

APPENDIX B
LABEL-ORDER CLASS DIAGNOSTICS

This appendix shows the per-class accuracy evolution of the two label-based orders, with one heatmap per seed (Fig. 5): each individual run specializes to a single class rather than learning a balanced classifier, and the selected class can differ between seeds, which a seed-averaged heatmap would hide.

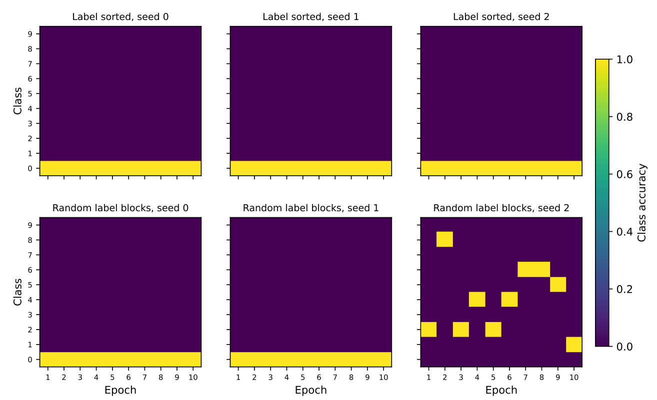

Fig. 5. Per-class test accuracy across epochs for the two label-based orders at learning rate 0.05, shown separately for each seed. Every run collapses onto a single class; `label_block_random` selects class 0 for seeds 0 and 1 but a different class for seed 2.

APPENDIX C
STABLE-METHOD DIAGNOSTICS

This appendix collects the gradient-alignment, class-balance, and forgetting diagnostics for the four trainable orders (Figs. 6–8).

REFERENCES

[1] L. Bottou, F. E. Curtis, and J. Nocedal, "Optimization methods for large-scale machine learning," *SIAM Review*, vol. 60, no. 2, pp. 223–311, 2018.

[2] M. Gürbüzbalaban, A. Ozdaglar, and P. A. Parrilo, "Why random reshuffling beats stochastic gradient descent," *Mathematical Programming*, vol. 186, pp. 49–84, 2021.

[3] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum learning," in *Proceedings of the 26th International Conference on Machine Learning*, 2009, pp. 41–48.

[4] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations*, 2019.

[5] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-mnist: A novel image dataset for benchmarking machine learning algorithms," *arXiv preprint arXiv:1708.07747*, 2017.

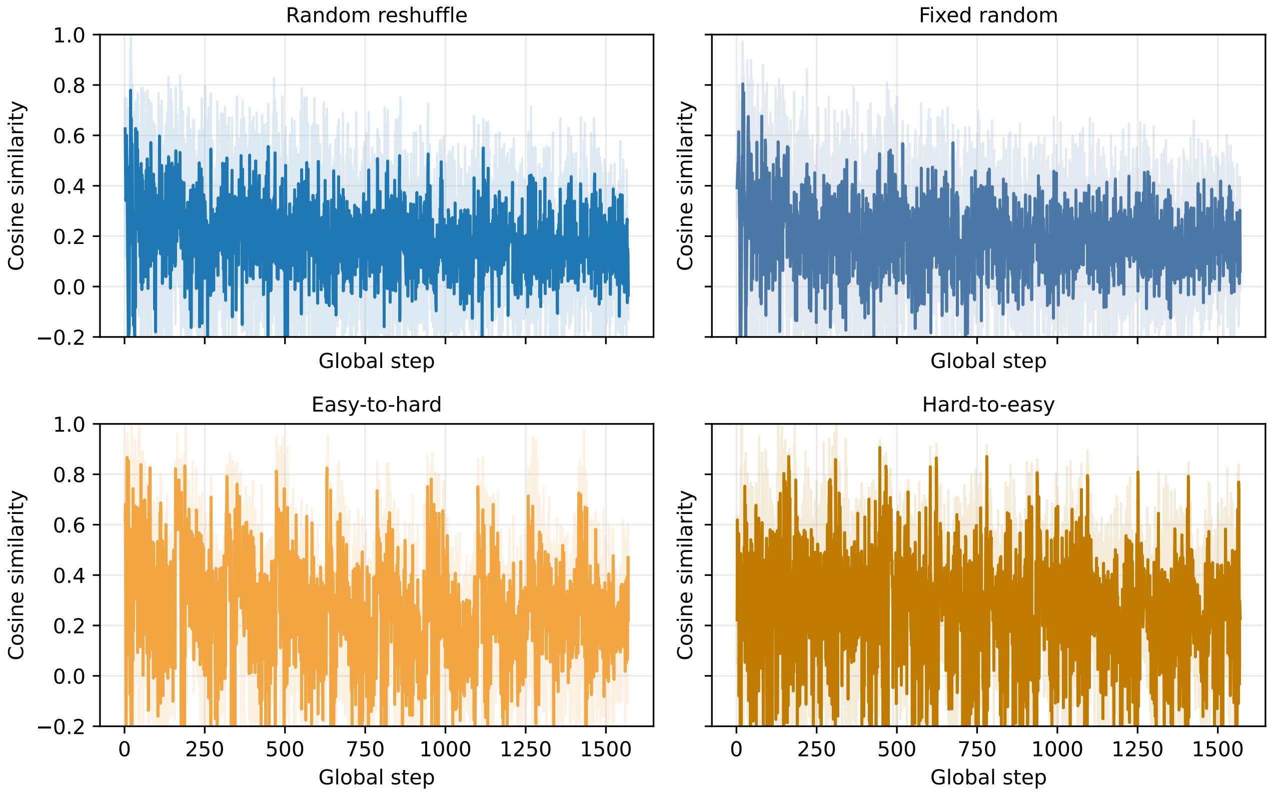

Fig. 6. Gradient cosine similarity for the four trainable orders, one panel per method. Shaded bands show one standard deviation across seeds.

[6] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," *Psychology of Learning and Motivation*, vol. 24, pp. 109–165, 1989.

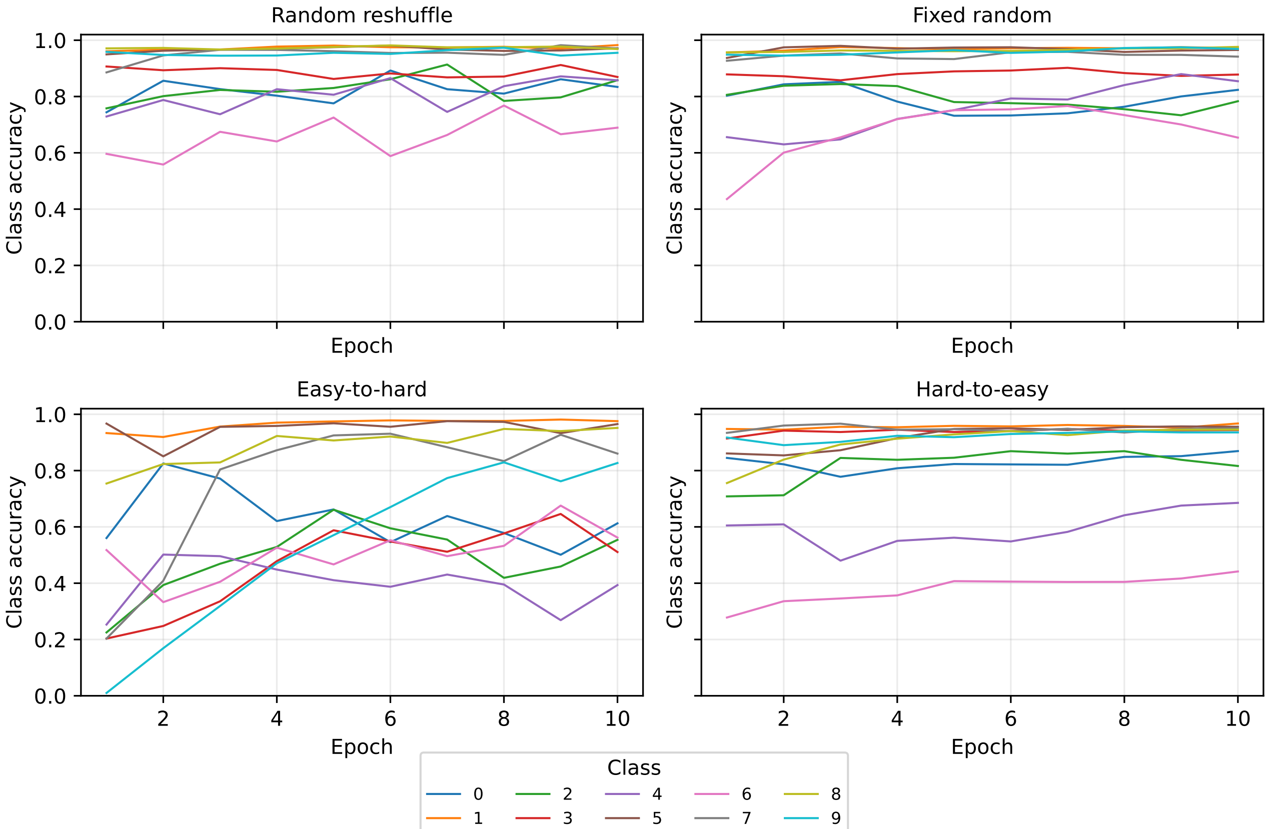

Fig. 7. Class-wise accuracy evolution for the four trainable orders. The curricula remain finite but produce less balanced class behavior than the IID baselines.

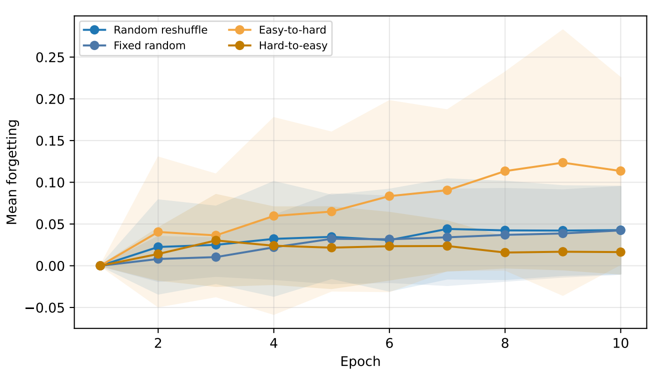

Fig. 8. Mean forgetting versus epoch for the four trainable orders. Easy-to-hard curriculum exhibits the largest retention gap.